

Linear Algebra for Machine Learning

Lecture Notes: Linear Independence and Rank

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🔗 Why This Matters for Math Students

As mathematics students, you are trained in the rigorous and abstract beauty of linear algebra. Concepts like vector spaces, span, and independence are pillars of pure mathematics. This lecture bridges that abstract world to the applied, data-driven field of Machine Learning (ML). You will see how these fundamental ideas are not just theoretical constructs, but essential tools for diagnosing problems in ML models (like multicollinearity), designing efficient algorithms (like PCA), and understanding the very structure of data itself. Your deep understanding of the "why" behind these concepts gives you a significant advantage in the world of data science and artificial intelligence.

1 The Building Blocks: Vectors, Linear Combinations, and Span

1.1 Vectors

A vector is an object that has both magnitude and direction. In the context of data, we can think of a vector as a list of numbers representing features. A vector in $\mathbb{R}^n$ is an ordered list of n real numbers.

Example A house can be represented by a vector $v = [\text{num_bedrooms}, \text{square_footage}, \text{age_in_years}]$. For a 3-bedroom, 1500 sq. ft. house that is 10 years old, the vector is:

$$v = \begin{bmatrix} 3 \\ 1500 \\ 10 \end{bmatrix} \in \mathbb{R}^3$$

1.2 Linear Combinations

A linear combination of a set of vectors $\{v_1, v_2, \dots, v_k\}$ is a new vector created by scaling each vector by a scalar (a real number) and adding the results.

💡 Formal Definition

A vector w is a **linear combination** of vectors $v_1, v_2, \dots, v_k$ if it can be written in the form:

$$w = c_1v_1 + c_2v_2 + \dots + c_kv_k$$

where $c_1, c_2, \dots, c_k$ are scalars.

Solved Example 1: Checking for a Linear Combination

Let $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. Is the vector $w = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ a linear combination of v_1 and v_2 ?

Solution:

We need to find if there exist scalars c_1 and c_2 such that $c_1 v_1 + c_2 v_2 = w$:

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

This vector equation translates into a system of linear equations:

$$1c_1 + 3c_2 = 5$$

$$2c_1 - 1c_2 = 3$$

From the second equation, we get $c_2 = 2c_1 - 3$. Substituting this into the first equation:

$$c_1 + 3(2c_1 - 3) = 5$$

$$c_1 + 6c_1 - 9 = 5$$

$$7c_1 = 14 \implies c_1 = 2$$

Now, we find $c_2 = 2(2) - 3 = 1$. Since we found a unique solution ($c_1 = 2, c_2 = 1$), **yes**, w is a linear combination of v_1 and v_2 . Specifically, $w = 2v_1 + 1v_2$.

1.3 Span

The **span** of a set of vectors $\{v_1, v_2, \dots, v_k\}$, denoted $\text{span}\{v_1, \dots, v_k\}$, is the set of *all possible* linear combinations of those vectors. It represents the entire space or subspace that can be "reached" using those vectors.

- The span of a single non-zero vector is a line through the origin.
- The span of two non-collinear vectors in $\mathbb{R}^3$ is a plane through the origin.

2 Linear Independence: The Concept of Redundancy

Linear independence answers the question: "Does my set of vectors contain any redundant information?"

2.1 Intuitive Definition

A set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others. If one vector *can* be written as a combination of the others, the set is **linearly dependent**.

- **Dependent (Redundant):** Let $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Here, v_3 is redundant because $v_3 = 2v_1 + 3v_2$. The set $\{v_1, v_2, v_3\}$ is linearly dependent.
- **Independent (No Redundancy):** The set $\{v_1, v_2\}$ is linearly independent because you cannot write v_1 as a multiple of v_2 , or vice-versa.

2.2 Formal Definition

💡 Formal Definition

A set of vectors $\{v_1, v_2, \dots, v_k\}$ is **linearly independent** if the only solution to the vector equation:

$$c_1v_1 + c_2v_2 + \dots + c_kv_k = \vec{0}$$

is the **trivial solution**, where $c_1 = c_2 = \dots = c_k = 0$.

If there exists any other solution (a **non-trivial solution**) where at least one c_i is not zero, then the set is **linearly dependent**.

2.3 How to Test for Linear Independence

1. Set up the homogeneous vector equation $c_1v_1 + \dots + c_kv_k = \vec{0}$.
2. Form an augmented matrix where the columns are your vectors, augmented with the zero vector.
3. Reduce the matrix to **Row Echelon Form (REF)** using Gaussian elimination.
4. Analyze the result:
 - **Independent:** If every column in the coefficient part of the REF has a pivot, the only solution is the trivial one.
 - **Dependent:** If there is a column without a pivot, there are free variables, leading to non-trivial solutions.

Solved Example 2: A Linearly Independent Set

Determine if the vectors $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent.

Solution:

We form the augmented matrix and reduce it.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right]$$

This REF matrix has a pivot in every column (columns 1, 2, and 3). This implies that the only solution is $c_1 = c_2 = c_3 = 0$. Therefore, the vectors are **linearly independent**.

Solved Example 3: A Linearly Dependent Set

Determine if the vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $v_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$ are linearly independent.

Solution:

We form the augmented matrix and reduce it.

$$\left[\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & 0 \\ 3 & 6 & 9 & 0 \end{array} \right] \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}]{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & -6 & -12 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The third column does not have a pivot. This means c_3 is a free variable, and non-trivial solutions exist. For instance, if we let $c_3 = 1$, we can back-solve to find $c_2 = -2$ and $c_1 = 1$. Since a non-trivial solution $(1, -2, 1)$ exists, the vectors are **linearly dependent**.

3 Rank of a Matrix

The rank of a matrix is a measure of its "information content" or the dimensionality of the space it represents.

3.1 Definitions of Rank

The rank of a matrix A , denoted $\text{rank}(A)$, can be defined in several equivalent ways:

1. The number of **linearly independent columns** of A .
2. The number of **linearly independent rows** of A . (Row Rank = Column Rank).
3. The dimension of the **column space** of A (the span of its columns).
4. The number of **pivots** in the Row Echelon Form of A . (This is the most practical way to calculate it).

3.2 Properties of Rank

Let A be an $m \times n$ matrix (m rows, n columns).

- $\text{rank}(A) \leq \min(m, n)$.
- **Full Rank:** A matrix has full rank if its rank is the maximum possible value, $\min(m, n)$.
- **Rank Deficient:** A matrix is rank deficient if $\text{rank}(A) < \min(m, n)$.
- An $n \times n$ square matrix is **invertible** if and only if it has full rank (i.e., $\text{rank}(A) = n$).

Solved Example 4: Finding the Rank

Find the rank of the matrix $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$.

Solution:

From Example 3, we found the Row Echelon Form of A:

$$\text{REF}(A) = \begin{bmatrix} \underline{1} & 4 & 7 \\ 0 & \underline{-3} & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

The pivots (first non-zero entry in each row) are underlined. There are **2 pivots**. Therefore, $\text{rank}(A) = 2$. This confirms that the 3 column vectors are linearly dependent and span a 2-dimensional subspace (a plane) in $\mathbb{R}^3$.

4 Introduction to Vector Coordinates and Basis Vectors

💡 Formal Definition

Key Idea: Vector coordinates can be understood as scalars that scale fundamental "basis" vectors, and the vector itself is the sum of these scaled basis vectors.

4.1 Understanding Coordinates Differently

Traditionally, a pair of numbers like $(3, -2)$ describes a vector by its x and y components. However, a more insightful way to think about these coordinates, central to linear algebra, is as **scalars** that stretch or squish other vectors.

4.2 The Standard Basis Vectors

In the familiar 2D Cartesian (xy) coordinate system, there are two special vectors:

- **i-hat ($\hat{i}$):** The unit vector in the x-direction. It points to the right and has a length of 1.
- **j-hat ($\hat{j}$):** The unit vector in the y-direction. It points straight up and has a length of 1.

These two vectors, $\hat{i}$ and $\hat{j}$, are together called the **basis** of the standard coordinate system.

Example: Consider the vector $(3, -2)$.

- The x-coordinate (3) scales $\hat{i}$ by a factor of 3, resulting in $3\hat{i}$.
- The y-coordinate (-2) scales $\hat{j}$ by a factor of -2 (flipping its direction and stretching it by a factor of 2), resulting in $-2\hat{j}$.
- The original vector $(3, -2)$ is then the sum of these two scaled vectors: $3\hat{i} + (-2)\hat{j}$.

This can be visualized as:

1. Scaling $\hat{i}$ to get $3\hat{i}$.
2. Scaling $\hat{j}$ to get $-2\hat{j}$.

3. Adding $3\hat{i}$ and $-2\hat{j}$ using the parallelogram rule (or tip-to-tail method) to get the resultant vector.

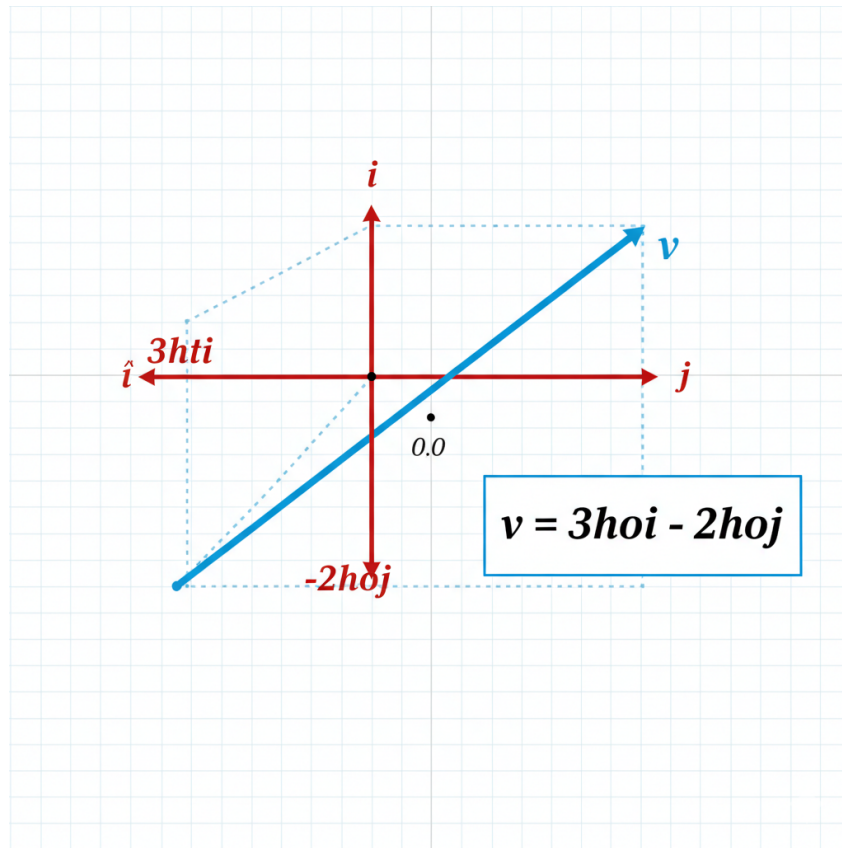


Figure 1: Visualizing a vector as a linear combination of basis vectors.

Concept Check: This way of thinking highlights that any vector can be expressed as a combination of these basis vectors.

4.3 Choosing Different Basis Vectors

The choice of $\hat{i}$ and $\hat{j}$ is standard, but **arbitrary**. We could choose any two non-aligned vectors as our basis vectors to create a new coordinate system.

Example: Imagine two new basis vectors, $\mathbf{v}_1$ (pointing up and to the right) and $\mathbf{v}_2$ (pointing down and to the right). Any 2D vector can still be reached by scaling $\mathbf{v}_1$ and $\mathbf{v}_2$ by some scalars (c_1, c_2) and adding them together: $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$. The specific pair of numbers (c_1, c_2) would now describe the vector in this *new* coordinate system.

Important Implication: Any time we describe vectors numerically (e.g., with coordinates), it implicitly depends on the chosen basis vectors.

5 Linear Combinations

Definition: A **linear combination** of two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is any vector formed by scaling each vector by a scalar and then adding the results. Mathematically, a linear combination is expressed as: $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, where c_1 and c_2 are scalars.

Why "Linear"? While not directly etymological, one way to think about "linear" is that if you fix one scalar (e.g., c_1) and let the other scalar (c_2) vary freely, the tip of the resulting vector ($c_1\mathbf{v}_1 + c_2\mathbf{v}_2$) will trace out a straight line.

Example: Let $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (our standard basis vectors).

- If we fix $c_1 = 2$ and let c_2 vary, the linear combination is $2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ c_2 \end{pmatrix}$. As c_2 changes, the tip of this vector traces out a vertical line at $x = 2$.
- If we fix $c_2 = 3$ and let c_1 vary, the linear combination is $c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ 3 \end{pmatrix}$. As c_1 changes, the tip of this vector traces out a horizontal line at $y = 3$.

6 Span

Definition: The **span** of a set of vectors is the collection of all possible vectors that can be reached by taking linear combinations of those vectors. It represents the "space" that the vectors can "generate" through scaling and addition.

6.1 Span of Two 2D Vectors

Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be two 2D vectors. There are two main possibilities for their span:

1. **Vectors do NOT line up (are not collinear):**

- In this common case, letting both scalars (c_1, c_2) range freely will allow you to reach **every possible point in the 2D plane**.
- The span is the **entire 2D space**.

2. **Vectors DO line up (are collinear):**

- If $\mathbf{v}_1$ and $\mathbf{v}_2$ point in the same direction or opposite directions, any linear combination $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ will still result in a vector whose tip lies on the **single line passing through the origin** that $\mathbf{v}_1$ and $\mathbf{v}_2$ define.
- The span is a **line** (a 1D subspace of 2D space).

6.2 Span of Two 3D Vectors

Consider two vectors, $\mathbf{v}_1$ and $\mathbf{v}_2$, in 3D space.

- If $\mathbf{v}_1$ and $\mathbf{v}_2$ do **not** line up, their linear combinations $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ will trace out a **flat plane** cutting through the origin of 3D space.
- This flat plane is the span of $\mathbf{v}_1$ and $\mathbf{v}_2$. It is a 2D subspace within 3D space.

7 Visualizing Collections of Vectors (Points vs. Arrows)

Guideline for Visualization:

- When thinking about a **single vector**, it's helpful to visualize it as an **arrow** with its tail at the origin.
- When dealing with a **collection of vectors** (like the span), it's often more convenient to represent each vector by its **tip as a point** in space. This prevents overcrowding the visualization with many arrows.

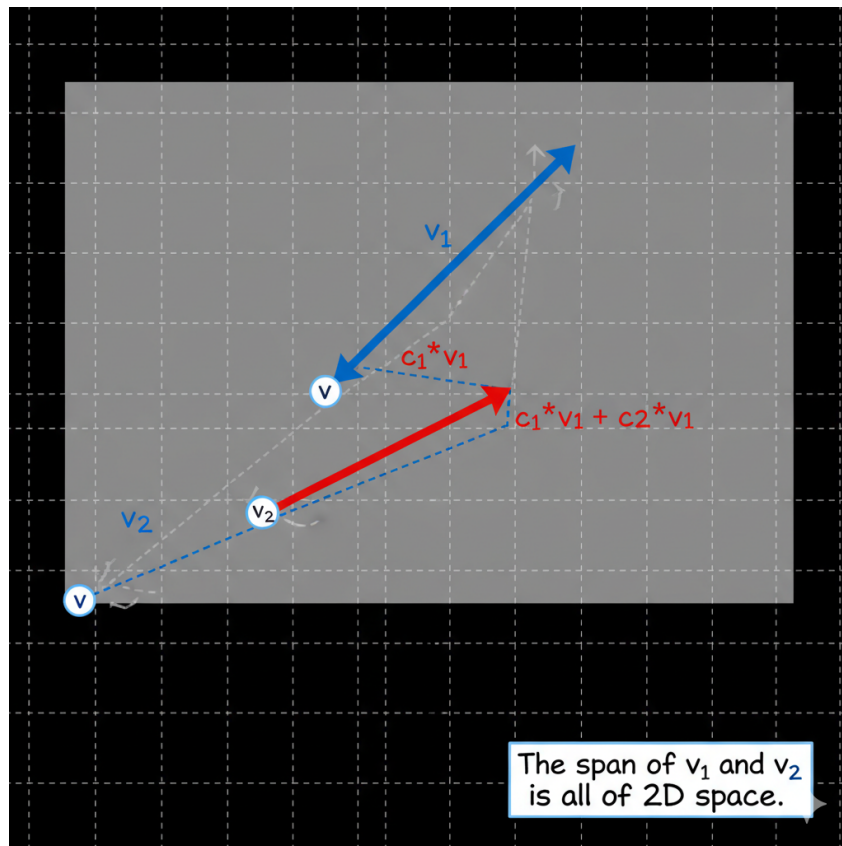


Figure 2: The span of two non-collinear 2D vectors is the entire 2D plane.

- The "line" representing the span of collinear 2D vectors is actually the set of all points that are the tips of those vectors.
- The "plane" representing the span of two non-collinear 3D vectors is actually the set of all points that are the tips of those vectors.

8 Span of Three 3D Vectors

Now, let's consider the span of three vectors, v_1 , v_2 , and v_3 , in 3D space. A linear combination of three vectors is $c_1v_1 + c_2v_2 + c_3v_3$.

Two main possibilities for their span:

1. Third vector IS on the span of the first two:

- If v_3 already lies on the plane formed by the span of v_1 and v_2 , then adding v_3 (even scaled) to the linear combination does **not** allow you to reach any new vectors outside that plane.
- The span remains the **same plane** as the span of v_1 and v_2 .
- The third vector is "redundant" in terms of expanding the span.

2. Third vector is NOT on the span of the first two:

- If v_3 points in a direction that is "out of" the plane formed by the span of v_1 and v_2 , then by scaling v_3 and adding it, you can "sweep" that plane through all of 3D space.

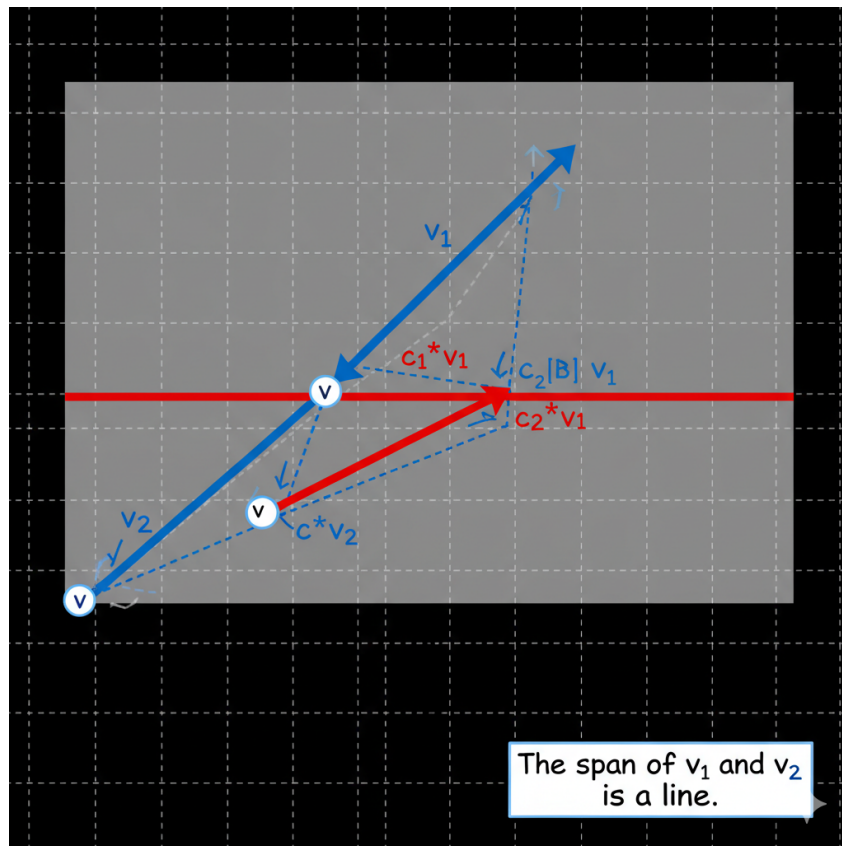


Figure 3: The span of two collinear 2D vectors is a line.

- The span becomes the **entire 3D space**.
- This means you can reach every possible 3D vector.

9 Linear Dependence and Independence

💡 Formal Definition

Key Idea: These terms describe whether a set of vectors is redundant in terms of expanding their span.

9.1 Linearly Dependent Vectors

Definition: A set of vectors is **linearly dependent** if at least one of the vectors can be removed without reducing the span of the set. Equivalently, one of the vectors can be expressed as a linear combination of the others (i.e., it already lies within the span of the other vectors).

Examples:

- In 2D, two vectors that line up (are collinear) are linearly dependent. If $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$, then $v_2 = 2v_1$. v_2 is a multiple of v_1 , so it's a linear combination of v_1 . Removing v_2 (or v_1) does not change the span (which is the line through the origin containing both vectors).

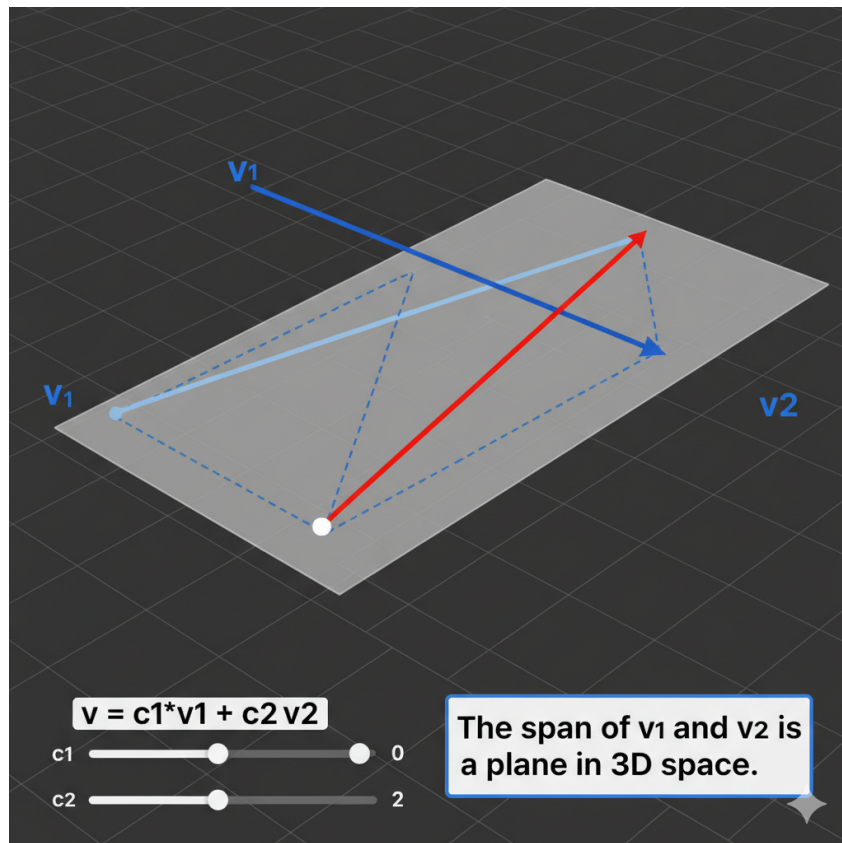


Figure 4: The span of two non-collinear 3D vectors is a plane in 3D space.

- In 3D, if three vectors (v_1, v_2, v_3) define a plane, and v_3 lies on the plane spanned by v_1 and v_2 , then they are linearly dependent. v_3 can be written as $c_1 v_1 + c_2 v_2$.

9.2 Linearly Independent Vectors

Definition: A set of vectors is **linearly independent** if each vector truly adds another dimension to the span. In other words, no vector in the set can be expressed as a linear combination of the others.

Examples:

- The standard basis vectors $\hat{i}$ and $\hat{j}$ in 2D are linearly independent, as $\hat{j}$ cannot be written as $c\hat{i}$ (and vice-versa). They span the entire 2D plane.
- In 3D, three vectors that do *not* lie on the same plane and point in genuinely different directions are linearly independent. They would span the entire 3D space.

10 The Technical Definition of a Basis

Definition: A **basis** of a vector space is a set of vectors that satisfies two conditions:

1. **They are linearly independent:** Each vector contributes uniquely to the span.
2. **They span the entire space:** Any vector in the space can be expressed as a linear combination of these basis vectors.

Why this definition makes sense:

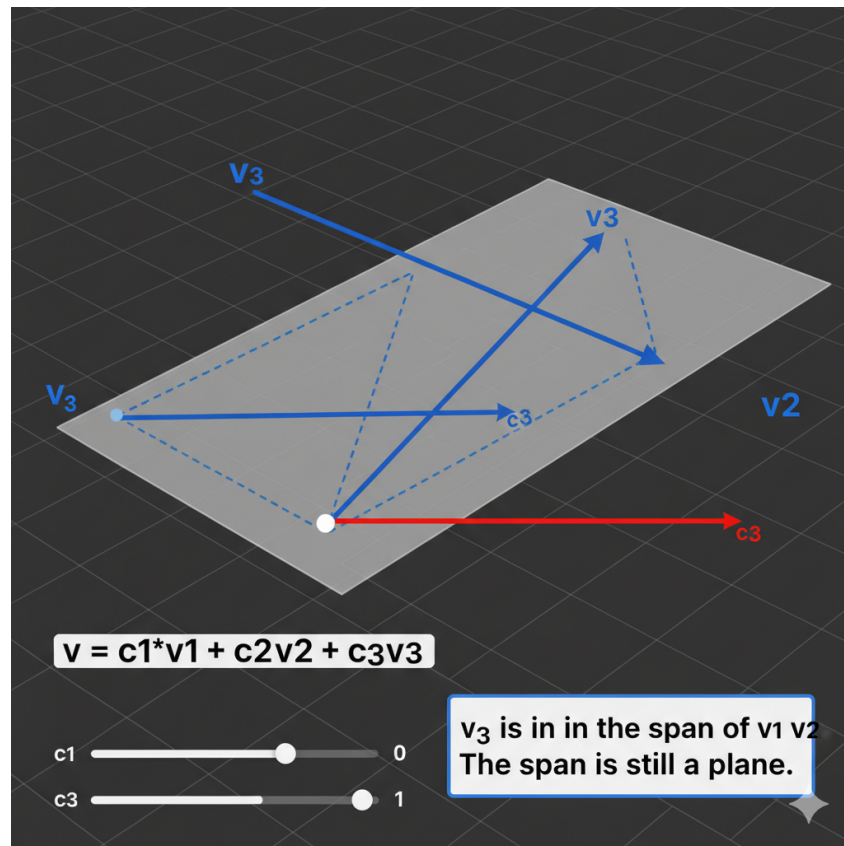


Figure 5: If the third vector is in the span of the first two, the span remains a plane.

- **Linearly Independent:** If the vectors were linearly dependent, it would mean one or more vectors are redundant. We could remove them, and the remaining (smaller) set would still span the same space. A basis should be an efficient set, not containing redundant vectors.
- **Span the Space:** The purpose of a basis is to provide a framework to describe *any* vector in that space. If they don't span the entire space, then there are vectors we cannot reach, which defeats the purpose of a coordinate system.

Therefore, a basis provides the minimal set of vectors required to "build" any other vector in the space using linear combinations.

11 Summary of Key Concepts

- **Vector Coordinates as Scalars:** Coordinates describe how much to scale basis vectors before adding them to form a given vector.
- **Standard Basis Vectors:** $\hat{i}$ and $\hat{j}$ are the standard unit vectors along the x and y axes, respectively, forming the standard basis for 2D space.
- **Linear Combination:** The sum of scaled vectors ($c_1 v_1 + c_2 v_2 + \dots$).
- **Span:** The set of all possible vectors that can be reached by taking linear combinations of a given set of vectors.
 - The span of two non-collinear 2D vectors is the entire 2D plane.

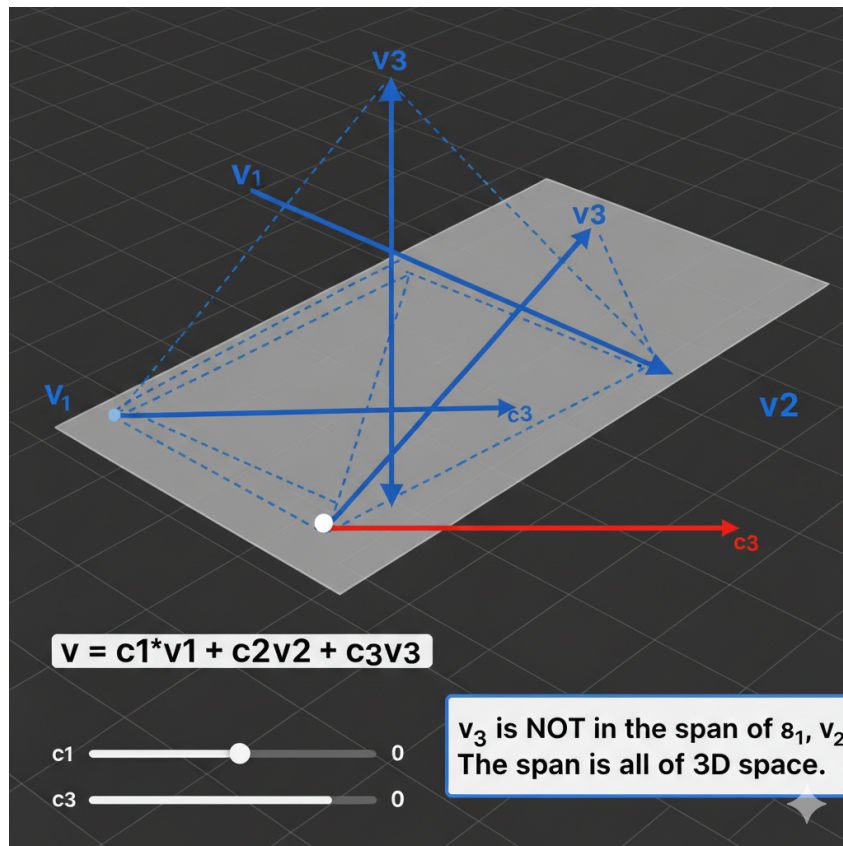


Figure 6: If the third vector is not in the span of the first two, the span becomes all of 3D space.

- The span of two collinear 2D vectors is a line.
- The span of two non-collinear 3D vectors is a plane.
- The span of three non-coplanar 3D vectors is the entire 3D space.
- **Linear Dependence:** A set of vectors where at least one vector can be expressed as a linear combination of the others (it is redundant in expanding the span).
- **Linear Independence:** A set of vectors where each vector genuinely adds a new dimension to the span (no vector can be expressed as a linear combination of the others).
- **Basis:** A set of linearly independent vectors that span the entire vector space. It's the most efficient set of vectors to describe all others in that space.

12 Exercise Problems

12.1 Exercise 1: Linear Combinations and Visualization

Given the vectors $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

- a) Calculate the linear combination $2u + 3v$.
- b) Calculate the linear combination $-1u + 0.5v$.

- c) What is the resulting vector if we fix the scalar for $\mathbf{u}$ as 1 and let the scalar for $\mathbf{v}$ vary? Describe the geometric shape formed by the tips of these vectors.

12.2 Exercise 2: Understanding Span in 2D

Consider the following pairs of vectors. For each pair, describe their span geometrically (a point, a line, or the entire 2D plane).

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

b) $\mathbf{c} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

c) $\mathbf{e} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \mathbf{f} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

d) $\mathbf{g} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$

12.3 Exercise 3: Linear Dependence and Independence

Determine whether the following sets of vectors are linearly dependent or linearly independent.

a) $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$

b) $\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{w}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

c) $\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

d) $\mathbf{y}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{y}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{y}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ (Hint: Does $\mathbf{y}_3$ lie on the plane spanned by $\mathbf{y}_1$ and $\mathbf{y}_2$?)

12.4 Exercise 4: The Basis Concept

Given the space $\mathbb{R}^2$ (the 2D plane):

- a) Are the vectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ a basis for $\mathbb{R}^2$? Justify your answer using the definitions of linear independence and span.
- b) Are the vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ a basis for $\mathbb{R}^2$? Why or why not?

13 Applications in Machine Learning

Machine Learning Applications

Linear independence and rank are not just abstract concepts; they have direct and critical implications in ML.

13.1 Multicollinearity in Linear Regression

Problem: In a linear regression model, $y = X\beta + \epsilon$, we find the coefficients β by solving the normal equations: $\beta = (X^T X)^{-1} X^T y$. Here, X is the feature matrix where each column is a feature vector.

Connection: If the columns of X are **linearly dependent**, this is called **multicollinearity**.

- This means one feature can be predicted from a linear combination of others (e.g., ‘height_i in feet’ is perfectly predictable from ‘height_i in inches’).
- Mathematically, if the columns of X are linearly dependent, X is not full column rank. This causes the matrix $X^T X$ to be **non-invertible** (singular).
- The inverse $(X^T X)^{-1}$ does not exist, and the equation for β cannot be solved directly. This leads to unstable models where small changes in the data cause wild fluctuations in the coefficients.

Solution: Check the rank of the feature matrix X . If it is less than the number of features, identify and remove redundant features. Alternatively, use regularization techniques like Ridge Regression.

13.2 Dimensionality Reduction with PCA

Problem: We have a dataset with many features (high-dimensional), and many of these features might be correlated and thus contain redundant information. We want to reduce the number of features without losing much information.

Connection: Principal Component Analysis (PCA) finds a lower-dimensional representation of the data by identifying new, linearly independent axes (principal components) that capture the maximum variance.

- The **rank of the data matrix** tells you the intrinsic dimensionality of your data. If you have 100 features but the rank is only 15, it means all your data points lie on a 15-dimensional subspace.
- PCA is a tool to find that subspace. By projecting the data onto the top k principal components (where $k = \text{rank}(X)$ or less), we create a **low-rank approximation** of the original data. This discards noise and keeps the core information.

13.3 Feature Engineering and Selection

When creating new features, it's vital to understand their relationship with existing ones. If a newly engineered feature is a linear combination of existing features, it adds no new information to a linear model and can cause multicollinearity. Calculating the rank of the feature matrix before and after adding a new feature can formally check if you've added new, independent information.

📌 Key Takeaways

- **Linear Combination:** The fundamental operation of scaling and adding vectors.
- **Span:** The set of all possible vectors reachable by linear combinations from a starting set.
- **Linear Independence:** A test for redundancy in a set of vectors. An independent set is efficient.
- **Rank:** A single number quantifying the number of linearly independent vectors in a matrix. It reveals the "true" dimension of the data or system.